

f:- True target function COL 774
OCT 14, 2025

$$Bias(x) = E_D [f(x) - M(x)] \quad x^u, y^u \sim \text{dist}$$

$$\# Var(x) = E_D [M(x) - \hat{M}(x)]^2 \quad \hookrightarrow \quad E_D [M(x)]$$

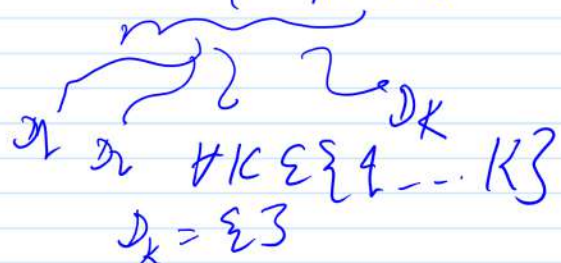
Net (squared) error = Bias² + Variance + Noise

Random Forests

collection of decision trees.

$$M_{\text{avg}}(x) = \frac{1}{K} \sum_{k=1}^K M_k(x) \quad \text{Data} = \{x^u, y^u\}_{i=1}^m$$

1. - 100
1, 3, 5, 7, ... 99



$\forall k \in \{1, \dots, K\}$
learn M_k on D_k

$$M_{\text{avg}}(x) = \frac{1}{K} \sum_{k=1}^K M_k(x)$$

① # of trees

②

$\{x_1, \dots, x_n\}$
 $\hookrightarrow$

($\sigma \subseteq n$)
 randomly choose σ sized subset
 S of variables to split on
 $\hookrightarrow$ choose best from S.
 (at every decision node)

$$\textcircled{3} E_k = \frac{1}{2} \sum_{i: (x^u, y^u) \notin D_k} [1 - \mathbb{1}_{M_k(x^u) = y^u}]$$